programming a 2-pin button

system 1 February 16, 2011, 12:28pm

I have a button(I will be more specific and post a picture if asked to) that has 2 outgoing pins
lets say I connect them to arduino uno's pin 7 and 8 what code should I use to read the button input?(like to check if the button is pressed)
any help would be much appriciated! XD

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robtillaart 2 February 16, 2011, 12:34pm

Please be more specific and post a picture (just kidding),

To get started you should use something like this. YOu can augment this by holding the state of every line so the printstatement is just printed once every press.

```
#define PIN1 7
#define PIN2 8

void setup()
{
  Serial.begin(9600);
  pinMode(PIN1, INPUT);
  pinMode(PIN2, INPUT);
  Serial.println("Start");
}

void loop()
{
  if (digitalRead(PIN1) == HIGH)
  {
    Serial.println("pin 1");
  }
  if (digitalRead(PIN2) == HIGH)
  {
    Serial.println("pin 2");
  }
}
```

A more complex if could be: if (digitalRead(PIN1) == HIGH && digitalRead(PIN2) == HIGH)

-- update

YOu must connect the common of the button to the +5V and use a pulldown resistor (10Kohm) between pin 7 and ground (idem for pin 8) to get things working.

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system 3 February 16, 2011, 12:38pm

First of all thanks for fast replying and it uses 2 pins but only one for data(I assume) so only one should matter but I can check that by myself
and does that mean that when a button is pressed its expressed as "HIGH" and when not pressed its "LOW" just like when a LED is on or off?

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robtillaart 4 February 16, 2011, 12:41pm

Maybe you better post the picture after all.

check - **http://arduino.cc/it/Tutorial/HomePage** - section 2 - there 50+ sample sketches there to learn how Arduino works. Time spent there is not lost !

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system 5 February 16, 2011, 12:48pm

I have the board the button and the resistor but not the breadboardd
can I somehow do that without the breadboard?
and what do you mean connect it to a 5v? how? and I thought that arduino gets 5v from the 5v battery/usb

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system 6 February 16, 2011, 12:50pm

> it uses 2 pins but only one for data

Buttons don't usually *do* data, other than the usual on/off - is this "button"device something like the one-wire memories sometimes used as ID tags in retail?

Like this:

Mixed-signal and digital signal processing ICs | Analog Devices

Analog Devices is a global leader in the design and manufacturing of analog, mixed signal, and DSP integrated circuits to help solve the toughest engineering challenges.

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robtillaart 7 February 16, 2011, 12:54pm

If you follow the scheme on - **http://arduino.cc/en/Tutorial/Button** - you don't necessary need the breadboard but put it on top of your shopping list!!
You can solder the wires or use ducttape or so; paperclips are also great tinkertoys

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system 8 February 16, 2011, 1:09pm

no it does not have any kind of serial number/ID on it
im attaching a picture of my arduino uno and button connected please tell me what I should do to make it work keep in mind that all I have is arduino uno, nano that button, leds and many kinds of bugs and transistors/etc

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system 9 February 16, 2011, 1:16pm

Wire it to ground and use the internal pull-ups.
No extra external components needed.

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system 10 February 16, 2011, 1:19pm

I know Im annoying but I just started dealing with arduino about a week ago...
what do you mean?
I connect one pin to lets say pin 7 and what about the other? connect it to a transistor on one side and the transistor to an arduino power pin on its other side?

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mmcp42 11 February 16, 2011, 1:22pm

noooo
one side of the switch direct to 5V
the other to pin 7
if you like you could put 100 ohm resistor in either wire

---

system 12 February 16, 2011, 1:25pm

how do I connect a pin to a 5v source?
I do have a 5v baterry but it has both red and black wires..and obviosly I would need to close the circuit...
so the main question would be how can I connect one pin to pin and one pin to a 5v source?

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system 13 February 16, 2011, 1:26pm

Noooo
One side of the switch direct to ground.
The other side to pin 7.
Enable the internal pull-ups - no need for external resistors.

system 14 February 16, 2011, 1:28pm

what does "direct to ground" mean here?
and how do I enable internal pull-ups?

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mmcp42 15 February 16, 2011, 1:29pm

connect one side of the switch to socket marked GND

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system 16 February 16, 2011, 1:36pm

now I get it
GND socket = the lower hole in a socket? but I thought it didnt do anything...
would it help? Is it dangerous?
before I do this can you give me a code that will show its working?
remember its a push down button not a regular one, which means it can give 2 values(I assume HIGH and LOW)
would this work?:

```
void setup()
{
Serial.begin(9600);
pinMode(7, INPUT);
pinMode(8, INPUT);
pinMode(13, OUTPUT);
Serial.println("Start");
}

void loop()
{

if (digitalRead(7) == HIGH)
{
digitalWrite(13, HIGH);
}
else if (digitalRead(7) == LOW)
{
```

```
digitalWrite(13, LOW);
}
else
{
}

}
```

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system 17 February 16, 2011, 1:37pm

You've been on the forum now for six days.
You've posted questions about quadratic equations, ASCII decimal arithmetic, C pointer basics, DC motor control, LCD screen, and now how to wire up a switch.

Have you thought about laying down some good foundations, and trying some of the tutorials and playground examples (which cover in detail just about all of your topics so far)?

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system 18 February 16, 2011, 1:39pm

no... After this I will do that so thanks for the suggestion
but please answer my questions and I will do what you suggested

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system 19 February 16, 2011, 1:43pm

> but I thought it didnt do anything...

It does - it completes the circuit (I know that wasn't a question, but you can have the answer for free)

> would it help?

Yes. See above.

> Is it dangerous?

Yes - the whole infrastructure of space-time will come crashing down around your ears.
(Only kidding)

```
if (digitalRead(7) == HIGH)
  {
   digitalWrite(13, HIGH);
  }
else  if (digitalRead(7) == LOW)
```

If it wasn't HIGH when you read it the first time, then there's no real need to read it again such a short time later - a simple "else" will suffice.

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system 20 February 16, 2011, 1:50pm

Thanks I'll try everything you said!
and I used the arduino digital button example for the code, I hope it works

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stephen_t 21 February 16, 2011, 1:57pm

The first picture on here of a knockoff Uno ?

Pity the real one doesn't run to crystals.....

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system 22 February 16, 2011, 2:20pm

> GND socket = the lower hole in a socket?

[Comprehension slowly dawns*] WOAH THERE!

No, not the earth connection on a mains socket.

- Here in the UK, the earth pin is at the top.

system 23 February 16, 2011, 3:14pm

THANK YOU THANK YOU THANK YOU!!!!!!!!!
instead of using an outlet I just connected one pin to pin 7 and the other to a power pin through a 10k ohm resistor!!!!

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system 24 February 16, 2011, 3:15pm

And im talking about an asian outlet(Israeli to be exact) I just did realize it differs

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system 25 February 16, 2011, 3:22pm

Im not sure whether this question worths a new topic or not but can I use a reistor to make a dc motor work with arduino
its pratically a 5v motor but I think it can work with less but It didnt work...
can an ohm resistor fix that too?

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system 26 February 16, 2011, 3:27pm

> its pratically a 5v motor but I think it can work with less but It didnt work...

You've already been told that you can't connect (most) DC motors directly to an Arduino.

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system 27 February 16, 2011, 3:31pm

Which is why im asking if a resistor would help - not direct

mmcp42 28 February 16, 2011, 3:34pm

I'm beginning to wonder if we have a troll in our midst

just take a look at your Arduino
don't connect it up
just look at it

do you see pins maked 5v? they will provide 5volts
do you see pins marked GND (GrouND) they are 0volts

when we say connect one side of the switch to ground - guess which pin we mean!

NEVER EVER EVER connect anything to the mains except a mains adaptor
no NEVER

did I mention NEVER

on the other hand - go grab yourself a Darwin award
(and it you're not sure what that is try google)
((and if you're not sure what that is, i refer you to my first comment!))

have fun!

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system 29 February 16, 2011, 3:44pm

If I have a 5v pin why cant I connect a dc motor directly??

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mmcp42 30 February 16, 2011, 3:46pm

because it can't supply enough current to drive (most) motors

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retrolefty 31 February 16, 2011, 3:48pm

> tomer1510:
> If I have a 5v pin why cant I connect a dc motor directly??

Because that +5vdc pin can only provide so much current. How much current does your motor draw at +5vdc applied voltage when under normal load?

You have around 400ma of maximum current capacity for that 5 volt pin for all external components.

Lefty

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system 32 February 16, 2011, 3:56pm

> Which is why im asking if a resistor would help - not direct

OK here's the answer you want to hear.

Yes, you can connect a small DC motor to an output pin of your Arduino via a resistor.
The resistor will limit the current flowing and prevent damage to your Arduino.
As an added bonus, the limited current will prevent the motor turning, thereby saving you from injury to any of your body parts in contact with the (otherwise) rotating parts of the motor, and it will therefore save wear and tear on the motor's bearings.

Happy with that answer?

Related topics

Topic	Replies	Activity
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Controlling motors with pushbuttons (2 pin)	15	May 5, 2021
arg help, pulse in, multple buttons and gnd	6	May 6, 2021
[Arduino UNO] Button state change	27	May 5, 2021
Connecting 2-pin button	14	May 6, 2021